Geophys. J. Int. (1996) **127**, 305–310

Palaeomagnetism of the Jurassic Asphaltkalk deposits, Holzen, northern Germany

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Accepted 1996 June 18. Received 1996 June 18; in original form 1996 March 11

SUMMARY

Palaeomagnetic and rock magnetic studies on the Kimmeridgian–Portlandian Asphaltkalk deposits of the Hils Syncline near Holzen in northern Germany suggest a connection between bitumen impregnation and magnetization in the limestones. A stable remanent magnetization is only found in the Asphaltkalk deposits, not in non-bitumen-impregnated limestones of the same age. The stable remanence was acquired after folding in this region, as revealed by a palaeomagnetic fold test. The timing of remanence acquisition is Late Cretaceous–early Tertiary, as derived from a comparison of the mean pole of the magnetic component with the European apparent polar wander path (APWP).

The decay pattern of the natural remanent magnetization (NRM) during alternating field (AF) and thermal demagnetizaton indicates that the magnetization is carried by magnetite. Isothermal remanent magnetization (IRM) acquisition and thermal decay measurements, as well as hysteresis parameters, are consistent with this interpretation. This study suggests that the creation of authigenic magnetite during the lower Tertiary in the Asphaltkalk deposits might coincide with the timing of hydrocarbon migration into the unit.

Key words: Asphaltkalk, chemical remanent magnetization, magnetite.

INTRODUCTION

Many studies have shown a link between hydrocarbon migration and magnetization in limestones (e.g. Elmore *et al.* 1987; McCabe, Sassen & Saffer 1987; Kilgore & Elmore 1989; Reynolds *et al.* 1990). It is believed that the geochemical conditions produced by hydrocarbon migration can cause the creation of authigenic magnetic phases (e.g. Elmore *et al.* 1993; Machel 1995). This relationship can be used as a means of dating hydrocarbon migration events by deriving the timing of remanence acquisition from palaeomagnetic investigations. In this study we report palaeomagnetic and rock magnetic results from bitumen-impregnated limestones in northern Germany. We present evidence for the connection between hydrocarbon migration and the acquisition of a remanent magnetization. Furthermore, we attempt to date the migration event by resolving the age of the magnetization.

GEOLOGICAL SETTING

The bitumen-impregnated limestones of the Asphaltkalk deposits are of Kimmeridgian and Portlandian age. The sampling sites are located on both limbs of the Hils Syncline in northern Germany (Fig. 1). The time of folding in the area is believed to be post-Neocomian. The Asphaltkalk consists of 13 layers of ooidal and skeletal wackestone and packstone separated by less-permeable marlstones and other carbonates.

Most total organic carbon (TOC) values are in the range of 2 to 4 per cent (Horsfield *et al.* 1991). The amount of total organic carbon present in these limestones is due to bitumen impregnation, the degree of which is believed to be mainly controlled by ooid versus carbonate matrix abundance and the number of fractures present in the rock (Horsfield *et al.* 1991). Horsfield *et al.* (1991) suggest that the deposits represent an exhumed and heavily altered multilayered petroleum accumulation. Biodegradation, water washing and non-microbial oxidation have severely altered the composition of the original Asphaltkalk petroleum. Approximately 85 per cent of the original mature petroleum is believed to have been removed by alteration processes.

The Posidonia Shale of Toarcian age, found stratigraphically at about 200 m below the Asphaltkalk, is believed to be the main source for the hydrocarbons (Horsfield *et al.* 1991). For nearby locations, Horsfield *et al.* (1991) report vitrinite reflectance values of 0.48 to 0.68 per cent for this unit. An estimate of temperature based on vitrinite reflectance values

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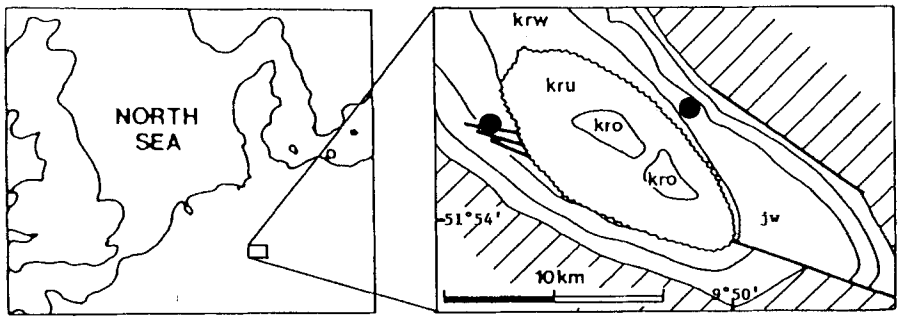


Figure 1. Location of the study area in northern Germany [Modified after Rullkötter *et al.* (1988) and Horsfield *et al.* (1991)]. Filled circles indicate sampling localities. jw: upper Jurassic; krw: Lower Cretaceous, Wealden; kru and kro, Upper and Lower Cretaceous.

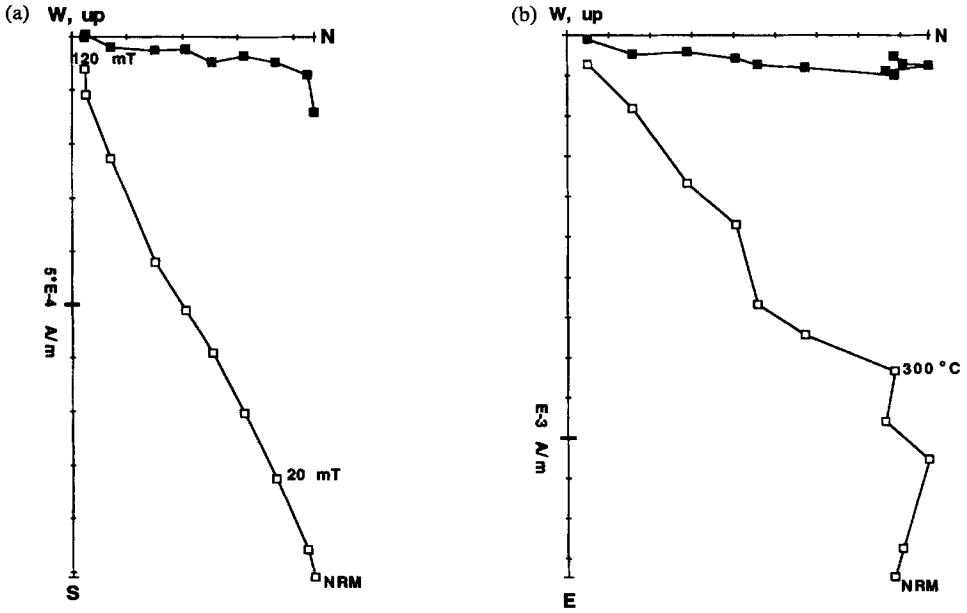


Figure 2. Representative orthogonal projections (*in situ*) showing decay during (a) AF and (b) thermal demagnetization. Solid squares indicate horizontal projection; open squares indicate vertical projection.

(Barker & Pawlewicz 1994) suggest that the Asphaltkalk was not heated beyond 120 °C.

Rullkötter *et al.* (1988) reported that significant petroleum generation and expulsion occurred in this region. Horsfield *et al.* (1991) indicate that the most likely timing of petroleum generation was during the Late Cretaceous (Turonian). The evidence for this interpretation is that the basin subsided to its maximum depth at that time and heat flow values were probably high due to a major igneous intrusion to the NW of this location.

METHODS

Palaeomagnetic samples were collected at several sites on both limbs of the Hils Syncline using a portable drill. They were oriented with a clinometer and a Brunton compass. The NRM was measured using 2G and United Scientific three-axis cryogenic magnetometers located in a magnetically shielded room. The specimens were demagnetized by stepwise alternating fields (AF demagnetization) or by heating in order to isolate the components of the NRM. Directional and intensity information was plotted on orthogonal projections (Zijderveld 1967) and line segments were selected to perform principal component anlalysis (Kirschvink 1980). The AF demagnetization proved to be more effective in removing the NRM. Thermal demagnetization was hampered by the acquisition of a viscous magnetization due to alteration of ferrous minerals above 500 to 600 °C. Mean directions were computed for each individual site before and after applying a structural correction. A fold test was performed to determine the best grouping of the site mean directions. These results were used to compute a palaeomagnetic pole for the component. The age of the magnetization was determined by locating this pole position on the APWP for Eurasia (Besse & Courtillot 1991).

The IRM acquisition and thermal decay measurements, in addition to intensity decay information during alternating-field experiments, were used to identify the magnetic minerals. Several samples from different sites were subjected to an IRM in fields up to 0.68 T prior to the thermal decay. Additionally, hysteresis experiments performed at the University of Utah contributed information about the magnetic characteristics of the bitumen-impregnated limestones.

RESULTS AND INTERPRETATION

At low AF strengths (< 20 mT) and low temperatures (< 200–300 °C) some samples of the Asphaltkalk deposits decayed

in irregular directions, whereas in other samples a northerly and steep-down component was removed. This component is interpreted as a modern viscous remanent magnetization. The NRM intensities of non-bitumen-impregnated limestones are substantially lower than those of bitumen-rich limestones. The mean magnetic intensity of non-impregnated rocks is $6.3 \times 10^{-5} \pm 3.14 \times 10^{-5}$ A m^{-1}, whereas the mean of all hydrocarbon-impregnated sites is $8.77 \times 10^{-4} \pm 5.92 \times 10^{-4}$ A m^{-1}. In addition to weak intensities, samples of non-impregnated limestones display poorly defined decay without consistent directions.

At high AF field strengths and temperatures, a second direction was removed in most hydrocarbon-impregnated samples. During AF demagnetization a northerly and downward-pointing component was usually removed between 20 and 90 mT (Figs 2a and 3). The same component is apparent during thermal demagnetization between 300 and 500 °C (Figs 2b and 3). Above 500 °C the specimens start to acquire a viscous magnetization due to the creation of a new magnetic phase, probably by oxidation processes at high temperatures.

The almost complete decay of the NRM during AF demagnetization and the removal of the magnetic direction below 500 °C suggest that magnetite is the carrier of the remanence. This conclusion is supported by IRM acquisition and thermal decay measurements (Fig. 4). The IRM acquisition reveals a low-coercivity phase and thermal decay suggests a maximum unblocking temperature at or above 500 °C. Hysteresis curves neither exhibit 'wasp-waisted' shapes (Fig. 5) nor do they have high values of remanent coercivity (H_{cr}) versus coercive force (H_c), which are frequently associated with remagnetizations (Channell & McCabe 1994). H_{cr}/H_c values between 1.82 and 2.78 and ratios of saturation remanence (M_{rs}) versus saturation magnetization (M_s) between 0.1 and 0.185 plot in the pseudo-single-domain field of Day, Fuller & Schmidt (1977). Examination of thin sections under reflected light showed the presence of an iron oxide associated with pyrite. This oxide could be magnetite that has replaced the pyrite (e.g. Suk, Pecor & Van der Voo 1990).

Site mean directions were calculated for all sites for *in situ* and bedding-corrected coordinates (Table 1 and Fig. 6). The site means were used to perform an incremental fold test to determine the timing of the magnetization relative to the folding event (Fig. 7). The fold test derived by McElhinny

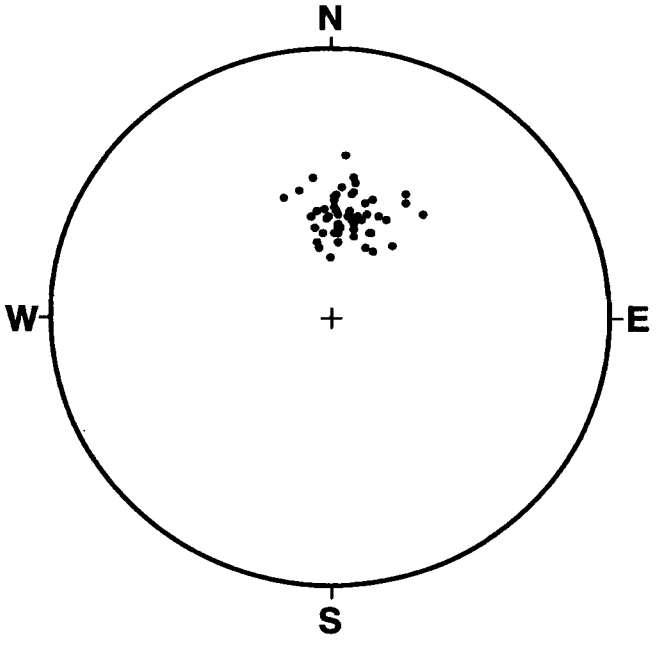


Figure 3. Equal-area projection of *in situ* specimen directions. Closed circles indicate a positive inclination.

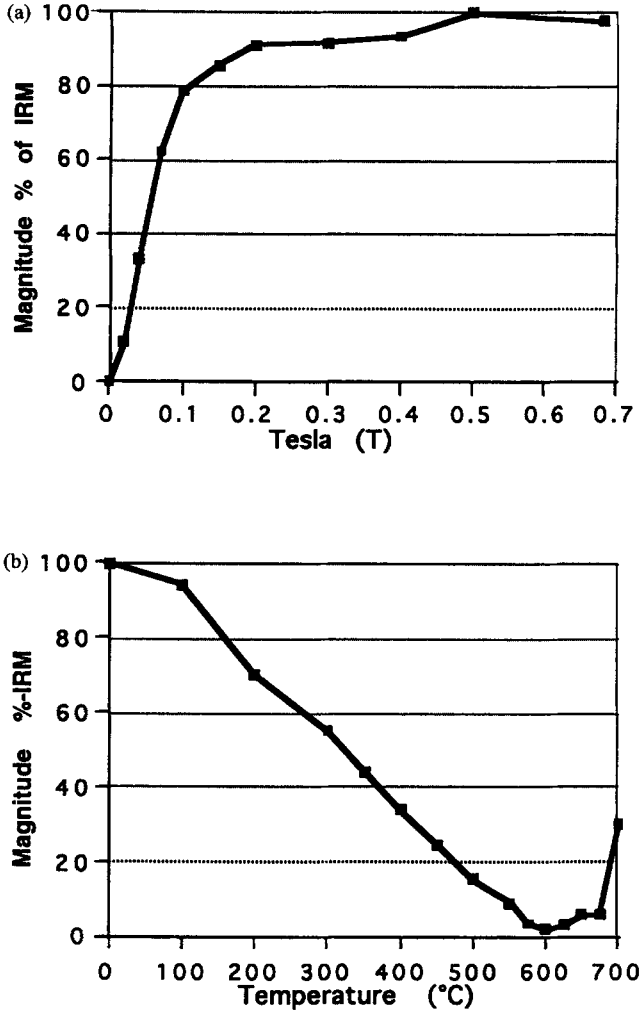


Figure 4. Magnetic characteristics of samples. (a) representative IRM acquisition curve; (b) representative thermal decay of IRM. The increase above 600 °C is due to the creation of new minerals.

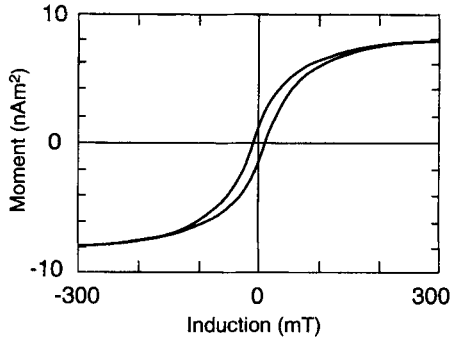


Figure 5. Representative hysteresis loop of hydrocarbon-impregnated limestone after diamagnetic slope correction.

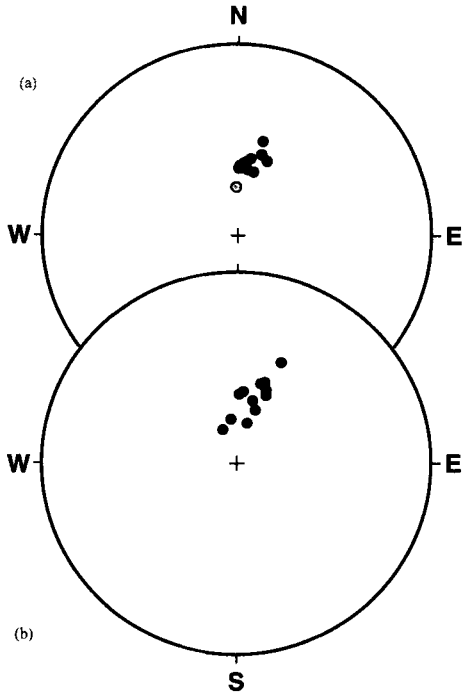


Figure 6. Equal-area projection of site mean directions. (a) *in situ* and (b) tilt-corrected directions. The shaded circle represents the Earth's present magnetic field (PEF).

(1964) showed a best grouping of the site means from both limbs at 6 per cent unfolding, but the differences between 6 per cent unfolding and *in situ* grouping are not statistically significant. The McFadden & Jones (1981) test showed a best grouping at 10 per cent unfolding. The fold test shows that the magnetization was not acquired prior to folding in this region, and the proximity of the best grouping to 0 per cent unfolding makes it most likely that the magnetization was acquired after folding. The mean magnetic direction at 0 per cent unfolding is at 9.2°E declination and 58.6° inclination, with a precision parameter (k) of 222 and an α_{95} of 2.9°. The coresponding palaeopole is located at 75.8°N latitude and 159.4°E longitude. Based on the pole position (Fig. 8) and a Cretaceous age for the folding, the magnetization was acquired during the Late Cretaceous–early Tertiary.

DISCUSSION AND SUMMARY

On the basis of the blocking-temperature–relaxation-time curves from Pullaiah *et al.* (1975), the burial termperatures of approximately 120 °C for the Asphaltkalk and the maximum unblocking temperatures of 500 °C suggest that the characteristic component is not thermoviscous in origin. The magnetization is interpreted as a chemical remanent magnetization (CRM). Hysteresis results are not consistent with parameters believed to be indicative of remagnetization (Channell & McCabe 1994).

The strong remanence in the bitumen-impregnated limestones, as opposed to the weak magnetization in non-impregnated samples, suggests a connection between the hydrocarbons and the magnetization in the Asphaltkalk deposits. The magnetization was acquired after folding in the area, excluding any mechanism directly triggered by deformation (e.g. strain). It has been shown previously that the creation of authigenic magnetite can be caused by hydrocarbon migration (e.g. Elmore *et al.* 1987). Therefore, we propose that the magnetization may date the migration event in this area. This age is consistent with that suggested for petroleum generation, i.e. Late Cretaceous (Horsfield *et al.* 1991).

ACKNOWLEDGMENTS

This study was supported by DOE grant DE-FG02-93ER14397 and a NATO grant for Collaborative Research (no. 062

Table 1. Palaeomagnetic results from the Asphaltkalk deposits of the Hils Syncline, northern Germany.

Site name	N/N_0	D *(in situ)*	I *(in situ)*	k	α 95	D (tilt)	I (tilt)	
NE-limb, quarry								
VG-1	9/10	22.1	56.3	110.4	4.9	14.2	72.7	
VG-2	5/6	3.8	59.5	124.6	6.9	337.5	74.6	
VG-3	5/6	9.5	56.8	160.9	6.0	333.8	71.1	
SW-limb, mine								
G-1	6/7	2.2	60.3	149	7.6	2.2	60.3	
G-2	5/5	4.8	59.3	126.5	8.2	4.8	59.3	
G-14	5/7	6.2	60.1	113.1	8.3	23.1	58.7	
G-23	5/6	0.9	61.6	390.1	3.9	19.5	65.9	
G-24	2/4	14.5	62.3	-	-	14.7	62.3	
G-25	7/8	16.2	54.0	41.4	9.5	16.2	54.0	
G-27	7/10	359.2	58.0	43.2	9.3	12.5	53.6	
G-28	2/11	15.4	48.4	-	-	23.2	42.0	
G-29	2/5	8.3	61.8	-	-	22.0	56.0	
						Pole Latitude	Pole Longitude	dp/dm
in situ mean	12/12	9.17	58.56	222	2.9	75.8	159.4	3.2/4.3

N/N_0: ratio of specimens or sites used in the statistical analysis to the number of samples studied; D: declination in degrees; I: inclination in degrees; k: statistical parameter (Fisher 1953) associated with the mean directions; α95: radius in degrees of the cone of 95 per cent confidence of the mean; dp and dm: semi-minor and semi-major axes in degrees, respectively, of the oval of 95 per cent confidence about the pole.

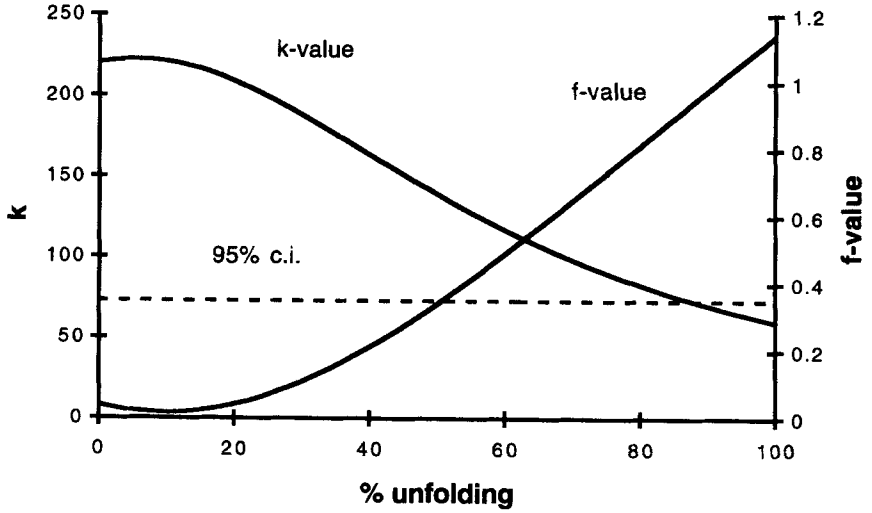


Figure 7. Incremental fold test. Precision parameter k and f value are plotted as a function of per cent unfolding. The dotted line indicates the 95 per cent confidence interval.

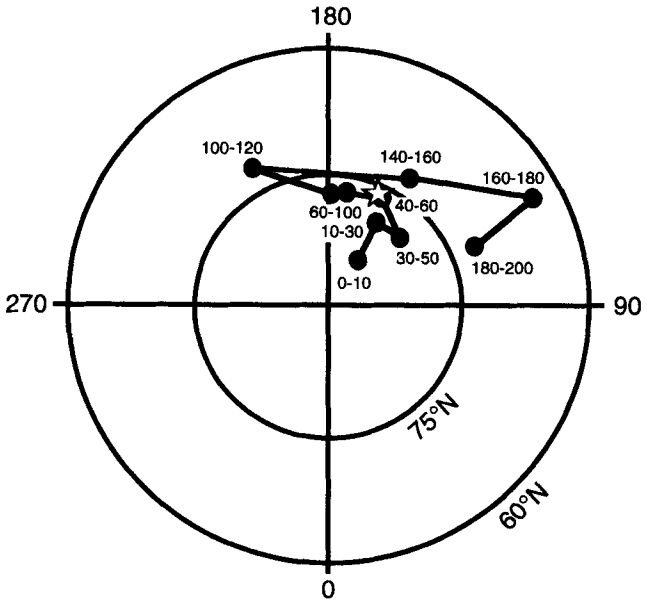


Figure 8. Eurasian apparent polar wander path [compiled data from Besse & Courtillot (1991)] with the palaeopole (asterisk) calculated from the *in situ* mean magnetic direction. Numbers on closed circles represent millions of years before present.

088LT12). We would like to thank J. Hedlers for his assistance during fieldwork.

REFERENCES

Barker, C.E. & Pawlewicz, M.J., 1994. Calculation of vitrinite reflectance from thermal histories and peak temperatures, in *Vitrinite Reflectance as a Maturity Parameter,* chapter 14, pp. 216–229, ed. Mukhopadhyay, P.K., American Chemical Society, Washington.

Besse, J. & Coutillot, V., 1991. Revised synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma, *J. geophys. Res.,* **96,** 4029–4059.

Channell, J.E.T. & McCabe, C., 1994. Comparison of magnetic hysteresis parameters of unremagnetized and remagnetized limestones, *J. geophys. Res.,* **99,** 4613–4623.

Day, R., Fuller, M. & Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and compositional dependence, *Phys. Earth planet Inter.,* **13,** 260–267.

Elmore, R.D., Engel, M., Crawford, L., Nick, K., Imbus, S. & Sofer, Z., 1987. Evidence for a relationship between hydrocarbons and authigenic magnetite, *Nature,* **325,** 428–430.

Elmore, R.D., Imbus, S., Engel, M. & Fruit, D., 1993. Hydrocarbons and magnetizations in magnetite, in *Applications of Palaeomagnetism to Sedimentary Geology, SEPM Special Publication No. 49,* pp. 181–191, eds Aïssaoui, D.M., McNeill, D.F. & Hurley, N.F., Tulsa, OK.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.,* A, **217,** 295–305.

Horsfield, B., Heckers, J., Lethaeuser, D., Littke, R. & Mann, U., 1991. A study of the Holzener Asphaltkalk, northern Germany: observations regarding the distribution, composition and origin of organic matter in an exhumed petroleum reservoir, *Mar. Petrol. Geol.,* **8,** 198–211.

Kilgore, B. & Elmore, R.D., 1989. A study of the relationship between hydrocarbon migration and precipitation of authigenic magnetic minerals in the Triassic Chugwater formation, southern Montana, *Geol. Soc. Am. Bull.,* **101,** 399–419.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Machel, H.G., 1995. Magnetic mineral assemblages and magnetic contrasts in diagenetic environments—with implications for studies of palaeomagnetism, hydrocarbon migration and exploration, in *Palaeomagnetic Applications in Hydrocarbon Exploration and Production, Geol. Soc. Spec. Pub. 98,* pp. 9–29, eds Turner, P. & Turner, A., The Geological Society, London.

McCabe, C., Sassen, R. & Saffer, B., 1987. Occurrence of secondary magnetite within biodegraded oil, *Geology,* **15,** 7–10.

McElhinny, M.W., 1964. Statistical significance of the fold test in palaeomagnetism, *Geophys J. R. astr. Soc.,* **8,** 338–340.

McFadden, P.L. & Jones, D.L., 1981. The fold test in Palaeomagnetism, *Geophys J. R. astr. Soc.,* **67,** 53–58.

Pullaiah, G., Irving, E., Buchan, K.L. & Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift, *Earth planet. Sci. Lett.,* **28,** 133–143.

Reynolds, R.L., Fishman, N.S., Wanty, R.B. & Goldhaber, M.B., 1990. Iron sulfide minerals at Cement Oil Field, Oklahoma: Implications for the magnetic detection of oil fields, *Geol. Soc. Am. Bull.*, **102,** 368–380.

Rullkötter, J. *et al.*, 1988. Organic matter maturation under the influence of a deep intrusive heat source: A natural experiment for quantitation of hydrocarbon generation and expulsion from a petroleum source rock (Toarcian shale, Northern Germany), in *Advances in Organic Geochemistry 1987*, pp. 847–856, eds Mattavelli, L. & Novelli, L., Pergamon Press, Oxford.

Suk, D., Pecor, D.R. & Van der Voo, R., 1990. Replacement of pyrite framboids by magnetite in limestone and implications for palaeomagnetism, *Nature*, **345,** 611–613.

Zijderveld, J.D.A., 1967. Demagnetization of rocks: Analysis and results, in *Methods in Palaeomagnetism*, pp. 254–286, ed. Collinson, D.W., Elsevier, New York.